# MILITARY SPECIFICATION

# SEPARATOR SYSTEM, OIL-WATER, COALESCER-TYPE, 5 GM CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a 5 gallon per minute (gpm) capacity, two-stage, coalescer-type, oil-water separator system with a prefilter stage and automatic oil discharge controls. The system shall include a supply pump and expendable coalescer and prefilter elements.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-S-781 | - Strapping, Steel, and Seals. |
| VV-F-800 | - Fuel Oil, Diesel. |
| MMM-A-1617 | - Adhesive, Rubber Base, General Purpose. |
| PPP-B-601 | - Boxes, Wood, Cleated-P_. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Comer. |
| PPP-B-636 | - Boxes, Shipping Fiberboard. |

Military

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| JAN-P-700 | - Paint, Deck, Gray, Inside. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIZ-D-1000 | - Drawings, Engineering and Associated Lists. |
| MIL-2104 | - Lubricating Oil, Internal Combustion Engine, Tactical Service. |

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| MIL-T-5624 | - Turbine Fuel, Aviation, Grades JP-4 and JP-5. |
| MIL-L-9000 | - Lubricating oil, Shipboard Internal Combustion Engine, High Output Diesel. |
| MIL-F-16884 | - Fuel Oil, Diesel, Marine. |
| MIL-S-52846 | - Separators, Oil-Water, Coalescer Type: Filter Elements, Fluid Pressure, Coalescer Oily Wastewater, Expendable: And Prefilter Elements, Oily Wastewater, Expendable. |
| MIL-F-52847 | - Filter Elements, Fluid Pressure, Coad.seer, Oily Wastewater, Expendable: And Refilter Elements, Oily Wastewater, Expendable. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-810 | Environmental Test Methods. |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of titivation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code; Section VIII, Pressure Vessels, Division 1.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, Ny 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION

Volume 5, Article 500 - Hazardous Locations.

(Application for copies should be addressed to the National. Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)



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AMERICAN SOCIETY FOR TESTING AND MATERIALS

B108 - Aluminum Alloy permanent Mold Casting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Roan 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC! 20036. )

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical. groups and using Federal agencies.

## 3. REQUIREMENTS

3.1 Description. The oil-water separator system (hereinafter called 'system") shall be a 5-gpm skid-mounted system consisting of three vessels - a prefilter vessel and two coalescer vessels connected in series (herein-after called “separator”); and a 5-gpm air-operated, double-diaphragm supply pump. The prefilter vessel shall utilize a single expendable prefilter element (hereinafter called a “refilter”), and the two coalescer vessels shall each utilize a single expendable filter-coalescer element (hereinafter called “elements”). The system shall be provided with an automatic oil discharge system connected to the prefilter vessel and the first coalescer vessel an automatic water discharge control valve on the system discharge, and automatic air eliminators installed on top of all three vessels. The system shall. also be equipped with four panel-mounted pressure gages to indicate pressure drop across each vessel. This panel shall also include sampling valves for obtaining samples from the system at the four points where the pressure taps for the gages are located, a rate-of-flow indicator



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connected to the system's water outlet, an “ON - OFF” power switch, with a red indicator light to indicate “01?” for the automatic oil discharge system; and, an air pressure regulator and gage for the air supply to the automatic oil and water control valves. Each vessel shall be equipped with sight gages that show the oil-water interface in the upper half of the vessels; and water drain valves on the bottom of each of the three vessels. The general requirements, operating, and environmental. conditions shall be in accordance with MIL-S-52846. The maximum work pressure of the system is 75 pound-s per square inch (psi).

3.1.1 Drawings. Unless otherwise specified (see 6.2), the proposor shall furnish as a part of his response to an invitation for bid or request for proposal a complete set of his preliminary engineering drawings and associated list (Category B, Form 3, MIL-D-1000) providing sufficient detail to enable evaluation and control of physical and functional design interrelationships of interdependent components, equipment, subsystems, systems, and controls. Drawings should show system configuration and all interface dimensional data applicable to the envelope, mounting, and mating of the assemblies and sub-systems l

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish a system for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce systems that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction exam-ination and tests, as specified (see 6.2).

3.3 Materials Materials shall be as specified herein or as specified on the Government approved engineering drawing and associated lists (see 3.1.1). Materials not specified shall be selected by the supplier and shall be sub-ject to all provisions of this specification.

3.3.1 supply pump When a supply pump is specified (see 6.2), electric-motor driven pumps shall be of all bronze construction with “K” monel pump and motor shaft, shaft key, and fasteners. Air operated pumps and surge suppressors shall be aluminum alloy conforming to ASTM B108 All robber wetted parts of the supply pumps shall be of nitrile rubber (NBR).



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3.3.2 Valves. Unless otherwise specified (see 6.2), all valves in the system except the pump pressure relief valve and check valves shall be full-flow ball type. Valves in the oil discharge line shall be steel or bronze. Valves in contact with water or oil-water mixtures shall be 316 stainless steel or bronze with 316 stainless steel trim.

3.4 Environmental conditions.

3,4.1 Operating temperature. The system, (separator, elements, prefilter, supply pump, and associated valves, piping, and controls), when assembled, shall permit rated flow of 5 gpm in any ambient temperature in the range of plus 125°F to plus 33°F.

3.4.2 Storage temperature. The system (separator, elements, prefilter, supply pump, and associated valves, piping, and controls), when assembled, shall not be damaged by storage at ambient temperatures from plus 160°F to minus 30°F, after which they shall operate as specified in 3.4.1.

3.5 Safety. All electrical components of the separator automatic control systems shall be explosionproof and classified as Class 1, Division I, Group D, in accordance with Article 500 of the National Fire Protection Association. All electrical systems shall be guarded and grounded to protect all persons and objects from electrical shock hazards.

3.6 Maintainability.

3.6.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication-charts as specified (see 6.6 and 6.7) shall be adequate to perform the intended function of assisting or conducting maintenance operations on the system.

3.6.2 Maintenance ratio. When specified (see 6.2), the system shall have a maintenance ratio of not greater than 0.03 when tested as specified in 4.5.2. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating the. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded.

3.7 Reliability. Unless otherwise specified (see 6.2), the specified Mean-The-Between-Failure (MTBF) shall be 140 hours when the systems are operated at 5 gpm and tested as specified in 4.5.2.



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3.8 Technical requirements. Unless otherwise specified (see 6.2), materials in the system shall not be adversely affected by and shall have no adverse affect on waters, from potable water to seawater, containing varying amounts of the following fuels and lubricants, alone or in combination:

| | |
|---|---|
| VV-F-800 | MIL-T-5624 |
| MIL-L-2104 | MIL-F-16884 |
| MIL-L-9000 | |

3.8.1 Separator vessels. Unless otherwise specified (see 6.2), the separator vessels, including the prefilter vessel, shall be desired and built in accordance with the requirements of the ASME Boiler and Pressure Vessel Code (Section VIII, Division 1) and need not be stamped. The pressure vessels shall withstand a hydrostatic pressure of 113 psi for a period of 5 minutes without any evidence of structural failure or permanent deformation.

3.8.2 Vessel viewports. When specified (see 6.2), each filter-coalescer vessel shall contain two 2-inch diameter tempered-glass viewports located 180 degrees apart, about 2 inches above the lower element mounting surface.

3.8.3 Automatic air vents. Unless otherwise specified (see 6.2), the three pressure vessels shall be equipped with automatic air venting devices in the quick-opening closure. Each venting device should be provided with a length of copper tubing or hose sufficiently long to reach ground or deck level, and brought together at a common point for collection of the vented fluids and vapors.

3.8.4 power requirements. Unless otherwise specified (see 6.2), the system, when equipped with an air-operated supply pump, shall require 117 V ac plus or minus 10 percent, 60 Hertz, single phase at 10 amperes or less; or 26.5 V dc plus or minus 10 percent (see 6.2). For electric-motor driven supply pumps, power requirements shall be 117 V ac plus or minus 10 percent, 60 Hertz, single phase; or 120 V dc and 26.5 V dc as specified (see 6.2). Power requirements shall not exceed 30 amperes for motor-driven pumps. The system with either type supply pump, shall require a compressed air supply up to 50 psi and one standard cubic foot per minute (sctm).

3,8.5 Physical requirements. Unless otherwise specified (see 6.2), the overall dimensions of the skid-mounted separator without the supply pump shall not exceed 64 inches long X 24 inches wide X 36 tithes high, Service clearance shall not exceed 12 inches in addition to the vertical dimension. All pressure vessels shall have a minumum outside diameter of 10 inches. For shipboard application each component of the system shall be capable of being passed through an opening 26 inches square without cutting or rewelding of the components.



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3.8.6 Automatic oil discharge system. Unless otherwise specified (see 6.2), the automatic oil discharge cycle shall be controlled by capacitance-type probes in the prefilter and first coalescer stages. The two oil discharge valves and the water discharge valve shall be solenoid-controlled air valves that can be adjusted to prevent instant opening and closing of the valves. All automatic valves in the oil discharge system shall be normally closed when the air supply or electrical power is turned off or otherwise fails.

3.8.7 Filter-coalescer elements. The filter-cod.ester elements shall be the standard expendable military element, Type IIII conforming to MIL-F-52847.

3.8.8 Prefilter elements. Unless otherwise specified (see 6.2), the prefilter elements shall be the standard expendable military prefilter, Type II (inside-to-outside flow), conforming to MIL-F-52847.

3.8.9 Identification marking. Unless otherwise specified (see 6.2), the separator shall be identified with an identification plate conforming to Figure 1, and attached to the prefilter vessel between the oil discharge line and the sight gage. The plates shall be attached to the vessel using adhesive conforming to MMM-A-1617, Type III.

3.8.10 Instruction plates. Unless otherwise specified (see 6.2), instruction plates conforming to Figure 2, and a warning plate conforming to Figure 3 shall be attached to the separator vessels as follows: “INLET” on the prefilter vessel directly above the inlet pipe; “OUTLET” on the second coalescer vessel directly above the outlet pipe; “WARNING” directly below the oil discharge valve on the first coalescer vessel; and “INSTRUCTIONS” plate directly below the oil discharge valve on the second coalescer vessel. The adhesive specified in 3.8.9 shall be used for attaching all plates to the vessels.

3.8.11 Pressure relief and check valves. Unless otherwise specified (see 6.2), a pressure relief valve shall be installed in the supply pump discharge line. The pressure relief valve shall be set to open at 50 plus or minus 5 psi. The bypass from the pressure relief valve shall. be plumbed into the suction side of the supply pump. Unless otherwise specified (see 6.2), the system shall also include check valves ahead of the supply pump, after the automatic water discharge valve, and at the end of the oil discharge Line on the second coalescer vessel.

3.8.12 Treatment and painting The portions of the system normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. Type B. The color shall be equipment gray in accordance with JAN-P-700.



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3.8.13 Government-furnished property. Unless otherwise specified herein, the quantity of filter-coalescer and prefilter elements conforming to MIL-F-52847, shall be furnished by the Government (see 6.5). When specified (see 6.2), the elements and prefilters required shall be furnished by the supplier and accepted by the Government based on successfully passing the applicable inspection requirements of MIL-F-52847.

3.8.14 cleaning and drying after tests. After conclusion of all tests, and removal of all used elements and prefilters, the system shall be wiped clean and air-dried at room temperature to exhaust all traces of water and test oils.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Comporient and material inspection. The supplier is responsible for insuring that components and materials used are manufacturee, examined, and tested in *accordance* with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model system shall be cause for disassembly, in the presence of a Government representative, of the preproduction model system to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, *tolerances, and* workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model system. Reassembly with replacement parts and retesting shall be the responsibility of the supplier. Government-furnished or Government-loaned property shall be excluded from the requirements of this paragraph.



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4.2 Classification of inspections. Inspections shall be classified as follows :

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Prepoduction inspection.

4.3.1 Examination. The preproduction system shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction system shall be tested as specified in 4.5.2.1 and 4.5.2.2 and when specified (see 6.2), 4.5.2.3, 4.5.2.4, and 4.5.2.5. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defect characteristics and 6.5 percent defective for minor defect characteristics.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 4 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The systems shall be examined as specified herein for the following defects:

Major

101. Components missing or not as specified.
102 l Dimensions not as specified.
103, Pressure vessels not coated as specified.
104 l Weldments not as specified.
105 Treatment and painting not as specified.
106. Leads in ressure vessels piping, valves, or components.
107. Workmanship not as specified.



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Minor

201. Identification marking incorrect, illegible, or missing,
202. Instruction plates not as specified.
203. Systems not cleaned and air-dried after testing.
204. Sealing plate gaskets missing or damaged.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified (see 6.2), the systems shall be tested at 5-gpm water flow rate. Other test conditions, test system arrangement, and test water shall be as specified in MIL-S-52846.

4.5.2.2 Performance. Unless otherwise specified (see 6.2), the system shall be tested to the following performance requirements as specified in MIL-S-52846:

(a) Hydrostatic pressure test, pressure vessels.
(b) Hydrostatic pressure test, inlet compartment.
(c) Differential pressure (first set of elements and prefilters),
(d) Oil removal (second and third hours only).
(e) Solids removal, coalescer elements only.
(f) Solids retention capacity, coalescer elements only.
(g) Solids removal, prefilter element only.
(h) Solids retention capacity, prefilter and prefilter elements only l

4.5.2.3 Operating and storage temperature. When specified (see 4.3.2), the systems shall be tested as specified in MIL-STD-810, Method 501, Procedure 1, for maximum temperatures; and Method 502, Procedure 1, for minimum temperatures. The operating periods shall be 1 hour or the operating time required to check for any component failure due to the test operating temperatures. Nonconformance to 3.4 or the applicable requirements of MIL-STD-810 shall consitute failure of either temperature test.

4.5.2.4 Maintainability.

4.5.2.4.1 Maintenance support evaluation. When specified (see 4.3.2), maintenance operations required during preproduction testing shall be acccomplished to determine conformance to 3.6.1. Failure of the support elements to maintain the system as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or MTBF.



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4.5.2.4.2 Maintenance ratio. When specified (see 4.3.2), the maintenance ratio shall be computed during preproduction testing. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.5 Reliability. When specified (see 4.3.2), the preproduction system(s) shall be tested as specified in MIL-S-52846, oil removal and life tests. This test cycle shall be repeated until 550 hours of operation have been accumulated. For testing purposes, a failure is defined as any malfuction which the operator cannot correct within 30 minutes by adjustment, repair, or replacement action using controls, on-equipment tools, or parts, and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels,
(b) Damage to the system by continued operation.
(c) A serious personnel safe by hazard.

(Note: Normal replacement of coalescer or prefilter elements shall not be considered a failure.) Occurance of six or more failures shall constitute failure of this test,

4.6 Inspection of preparation for delivery.

4.6,1 Quality conformance inspection of pack.

4.6.1.1 Unit of product, For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | |
|---|---|
| 108. | Materials, methods, and containers not as specified. Each incorrect material, method, or container shall be considered one defect. |
| 109. | Prefilter and elements not instilled as specified for Level A. |
| 110. | Drain valves not left open for Level A. |
| 111. | Openings not sealed as specified for Level A. |
| 112. | Supply pump not packaged as specified for Level A. |
| 113 l | Extra elements and prefilters not preserved and accordance with the referenced document for Level A |
| 114 . | Instructional material not secured to the separator as specified |
| 115. | Spare parts 'not preserved and packaged as specified for Level A. |
| 116. | Packing not as specified for Level A or B. |
| 117 . | Marking missing, illegible, incorrect, or incomplete. |



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## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Separator system. The separator system with the prefilter and elements installed in the separator shall not require preservation. Drain valves shall. be opened. ALL openings into the system, including drain valves, shall be sealed with metal, plastic, or fiber, dust plugs or caps as applicable.

5.1.1.2 Supply pump. When a supply pump is specified (see 3.3.1), it shall be packaged in a close-fitting box conforming to PPP-B-636, W5c, style optional. Fiberboard pads, constructed of the same material as the box shall be provided within the box to prevent movement or damage to the pump. The box shall be closed and sealed as specified for Method V in the appendix to the box specification.

5.1.1.3 Extra elements and prefilters, When extra elements and prefilters are furnished (see 3.8.13) they shall. be preserved and packaged in accordance with Level A requirements of MIL-F-52847.

5.1.1.4 Instructional material.. Instructional material shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3 and secured to the separator.

5.1.1.5 Spare parts. The preservative application criteria and applicable method(s of preservation of MIL-P-116 shall be used to preserve spare parts. The parts shall then be packaged together in a close-fitting box conforming to PPP-B-636, W5c or W6c as applicable to the weight of contents.

5.1.2 Level c. Each complete separator system shall be preserved and packaged in a manner to assure protection against deterioration and damage from the supplier to the initial. destination.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each complete separator system shall be packed in a close-fitting box conform to PPP-B-621, Class 2, style optional or PFP-B-601, Overseas Type, Grade B, style optional. The contents shall be blocked and braced within the box in a manner to prevent movement or damage. Box closure



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shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable, and unless otherwise specified herein, shall be Finish B. When specified (see 6.2), Finish A strapping shall be used.

5.2.2 Level B. Each complete separator system shall be packed as specified in 5.2.1 for Level A except boxes shall be Domestic Type or Class as applicable and box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level c. Each complete separator system shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in a container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The 5-gpm separator systems covered by this specification are intended for use aboard Army watercraft for removing the oil from oily bilge wastewater to meet oil pollution abatement requirements,

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When preliminary engineering drawings are required (see 3.1.1).
(c) Time frame required for submission of preproduction model (see 3.2.1).
(d) When the Government will conduct any or all of the preproduction examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2.1).
(e) When a supply pump is required, and what type of pump is required (see 3.3.1).
(f) When valves other than as specified are required (see 3.3.2).
(g) When maintenance ratio is required (see 3.6.2).
(h) When Mean-Time-Between-Failure for reliability other than as specified is required (see 3.7).
(i) When materials are to be compatible with fuels, lubricants, and waters other than those listed in 3.8.
(j) When vessels built to ASME pressure vessel code are not required (see 3.8.1).



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(k) When vessel viewports are required (see 3.8.2).
(l) When automatic air vents other than as specified are required (see 3.8.3).
(m) Power required for system (see 3.8.4).
(n) Power required for electric-motor driven supply pumps (see 3.8.4).
(o) When physical requirements other than those listed in 3.8.5 are required.
(p) Type of automatic oil discharge system required when other than as specified (see 3.8.6).
(q) Type of prefilter element required when other than as specified (see 3.8.8).
(r) Type of identification marking required when other than as specified (see 3.8.9).
(s) Type of instruction plates required when other than as specified (see 3.8.10).
(t) When pressure relief and check valves are not required (see 3.8.11.
(u) When the coalescer and prefilter elements are not to be Government-furnished (qualified products) (see 6.4), and are to be furnished by the supplier (see 3.8.13).
(v) When testing as specified in 4.5.2.3, 4.5.2.4, and 4.5.2.5 is required (see 4.3.2).
(w) When test conditions other than as specified in 4.5.2.1 are required.
(x) When performance tests other than as specified in 4.5.2.2 are required.
(y) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(z) When Finish A strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production systems from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish systems conforming to this specification.

6.4 Qualification. When the systems covered by this specification includes qualified components (coalescer and prefilter elements) (see 6.2), the contracting officer should include the requirements of ASPR 1-1107.2(b) in the solicitation.

6.5 Government-furnished property. When specified (see 6.2), the contracting officer should arrange to furnish the property specified in 3.8.13.



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6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each system,

6.7 Provisioning. The contracting officer should include provisioning requirements for spare parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of the systems.

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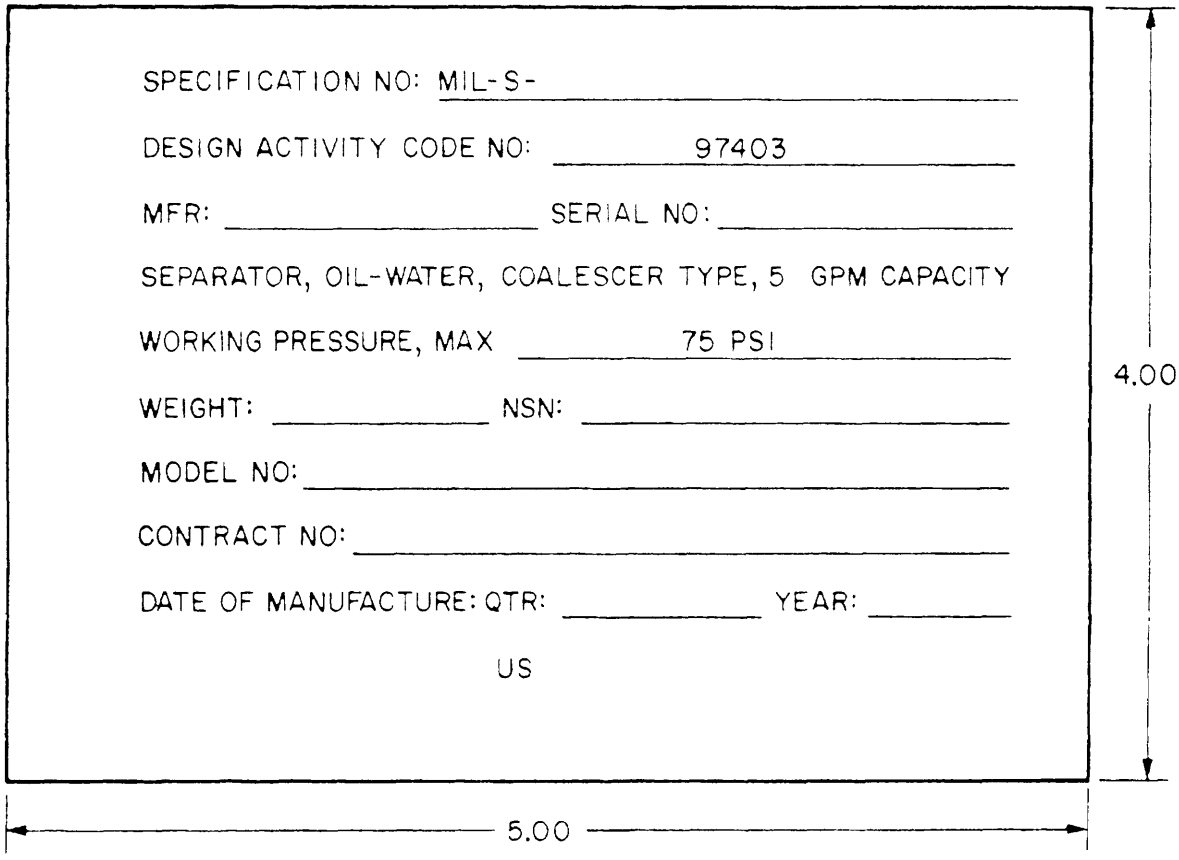
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Project No. 4330 -A150



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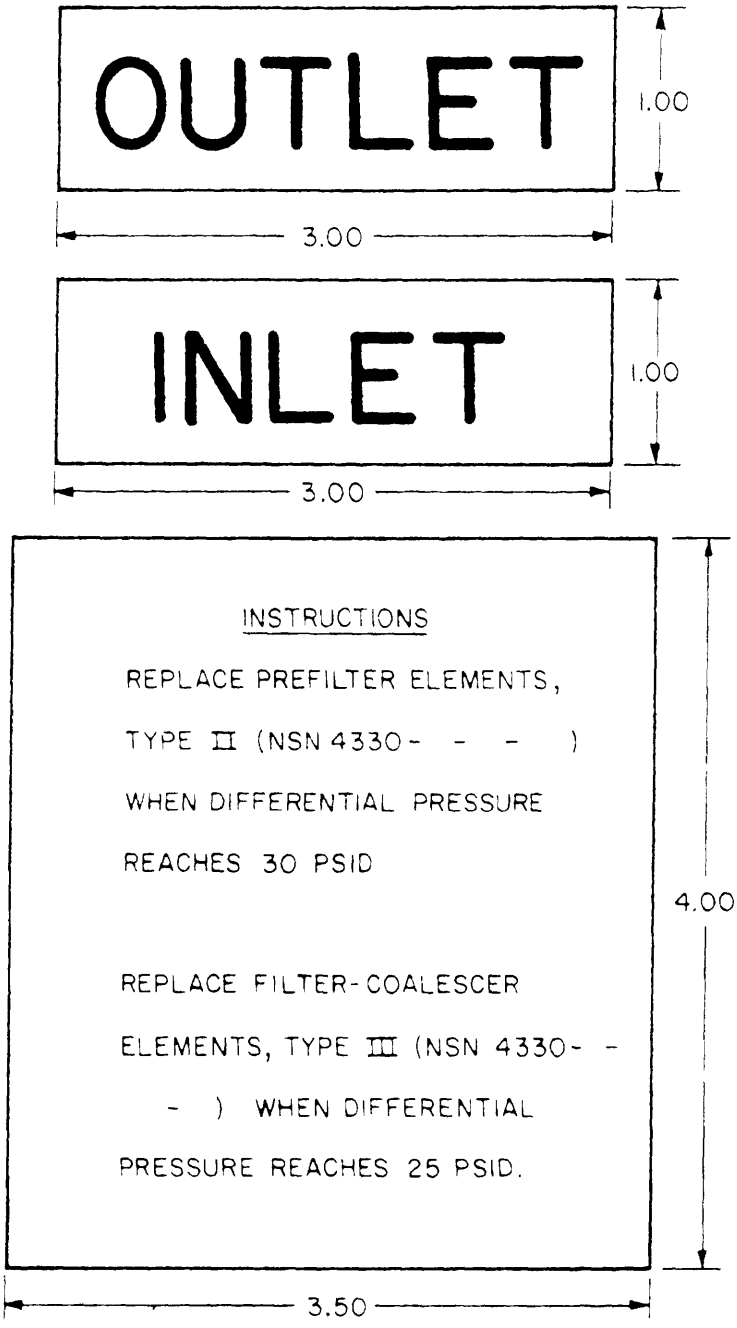

NOTES
1. DIMENSIONS ARE IN INCHES.
2 TOLEREANCE SHALL BE ± .06 .
3 MATERIAL SHALL CONFORM TO MIL-P-514, TYPE I, STYLE 1, COMPOSITION C, OF TYPE I, GRADE A, CLASS 1.

# FIGURE I. IDENTIFICATION PLATE

X-3035



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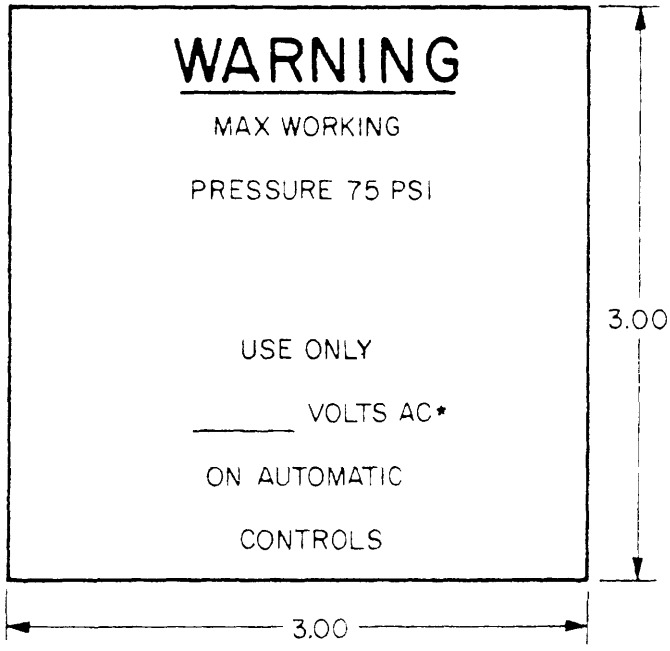



NOTES 1 DIMENSIONS ARE IN INCHES.
2 TOLEREANCE SHALL BE ± .06
3 MATERIAL SHALL CONFORM TO MIL-P-514, TYPE III, COMPOSITION C, OF TYPE I, GRADE A, CLASS 1.

FIGURE 2. INSTRUCTION PLATES X-3036



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* OR DC, AS APPLICABLE

NOTES: 1 DIMENSIONS ARE IN INCHES.
2 TOLEREANCE SHALL BE ± .06 .
3 MATERIAL SHALL CONFORM TO MIL-P-514, TYPE III, COMPOSITION C, OF TYPE I, GRADE A, CLASS 1.

FIGURE 3. WARNING PLATE

X-3037



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-S-52851(ME) Separator System, Oil-Water, Coalescer-Type, 5 GPM Capacity

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO |
|---|---|
| | DATE |

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED



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